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Divergent responses to climate change and disturbance drive recruitment patterns underlying latitudinal shifts of tree species

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Abstract

1. Climate change is expected to result in a reorganization of the continental distribution of tree species. Recent shifts in distribution patterns have been reported, but it is not always clear how climate change influences these patterns locally, especially in relation to other disturbances.
2. We investigated latitudinal shifts of four ecologically important tree species between 1970 and 2014 within a study area that encompasses their northernmost range limit in northeastern North America (Quebec, Canada; ~761,000 km²). Changes in latitudinal limits were defined in relation to changes in tree saplings' occurrence patterns within forest plots resampled over two time periods (1970–1977 and 2003–2014). By examining changes in the frequency of occurrences in different portions of the study area along a latitudinal gradient, we were able to identify spatially explicit patterns of loss or gain (sapling recruitment) resulting in the shifts observed. We then estimated the probability of observing a recruitment event in response to changes in climate, disturbance and their interaction, using a multimodel selection approach.
3. Latitudinal limits of all four species shifted northward, but these shifts resulted from different patterns of plot occurrence changes, depending on the species and the location examined. Greater recruitment at northern locations than at southern ones drove shifts for *Acer saccharum* Marsh., *Fagus grandifolia* Ehrh. and *Acer rubrum* L., but less so for *Betula alleghaniensis* Britt. Climate variables indicating changes in early or late growing season conditions were most often selected in models. Warming tended to reduce recruitment probability in the south but increase it in the north, leading to divergent responses for a given species across the study area. Disturbance effects were generally less important than climate change effects, as was their interaction.
4. **Synthesis.** Spatially explicit and divergent responses to climate change and disturbance drive recruitment patterns underlying latitudinal shifts of tree species. The importance of early- or late-season climate variables points towards biological processes being affected at critical stages of the life cycle. Understanding the factors that influence species' migration capacity in a changing climate is crucial to inform adaptive management and conservation practices.

KEYWORDS

climate change, forest dynamics, global change ecology, latitudinal shift, long-term forest monitoring, Quebec, Canada, range shift, tree migration

1 | INTRODUCTION

As climate change intensifies, the long-term consequences on biodiversity must be considered to minimize negative impacts on ecosystem productivity and livelihoods. Expected biotic responses to climate change include upward (elevational) or poleward (latitudinal) migration of species, although the former has been documented for plants more than the latter (Lenoir & Svenning, 2015; Yalcin & Leroux, 2017). Climate tends to change over larger distances along latitudinal gradients than for elevational ones, and tree range displacement involves relatively slow processes of reproduction, dispersal and establishment (Aubin et al., 2018). The magnitude of climate warming is particularly strong at northern latitudes and can put pressure on species to migrate towards more suitable habitats (Chen, Hill, Ohlemüller, Roy, & Thomas, 2011; Périé & de Blois, 2016; Xu et al., 2013). The pressure to shift could result in a long-term reorganization of the continental distribution of species in response to warming, with major consequences for the management of biodiversity and related regulatory frameworks (Berteaux et al., 2018).

For trees, evidence of potential latitudinal (or longitudinal) range shifts is mostly based on the observation of temporal changes in the spatial distribution of occurrence or abundance patterns (Boisvert-Marsh, Périé, & Blois, 2014; Fei et al., 2017; Sittaro, Paquette, Messier, & Nock, 2017; Zhu, Woodall, & Clark, 2012). Range shifts can result from a variety of site- and species-specific processes that lead to new occurrences or increased abundance at some locations and declines at others. For instance, a northward shift in latitudinal range limits for a species can result from increased recruitment in the northern portion of the range and/or from decreased site occurrence in the south (Lenoir & Svenning, 2015). A better understanding of changing distribution patterns in different portions of the range can therefore reveal important processes driving species responses at broad scales.

Climate change has been linked to range shifts of tree species during the Holocene (Davis & Shaw, 2001; Ordóñez & Williams, 2013), with site factors shown to influence local responses (Schwörer, Gavin, Walker, & Hu, 2017). While climate factors are also assumed to drive recent observations of range shifts (Evans & Brown, 2017), there is limited evidence of a direct relationship between patterns of range shifts and climate change. The availability of suitable data, such as repeated surveys of forest trees of North America over time periods at which climate change is usually considered (~30 years) and from longer chronologies established from tree rings, has begun to provide valuable insights into the spatially explicit responses of species to climate. For instance, whereas poleward shifts driven by increases in temperature were expected at high latitudes, westward shifts in species abundance associated with changes in moisture

availability were also documented for trees in the eastern United States (Fei et al., 2017). These westward shifts were larger (median rate 15.4 km per decade) than the latitudinal ones (11.0 km per decade) and resulted from increased stem density at the leading edge. The rate and direction of range shifts were shown to vary by region and species traits. Over the past ~60 years in boreal forests of Canada, tree growth rates demonstrated strong region- and species-specific trends, linking negative growth responses to increasing temperature and positive responses to increasing precipitation, but no consistent boreal-wide response to climate change (Girardin et al., 2016). Interestingly, in eastern North America, correlations between *Picea mariana* (Mill.) B.S.P. growth and temperature in Quebec, Canada differed across the study area, from negative south of 49°N to positive north of that latitude (D'Orangeville et al., 2016). Evidence from these recent studies points to spatially explicit processes that control tree growth or abundance and which cannot easily be averaged over the entire range of a species. Studies on the response of tree growth or abundance to changes in climate provide valuable information within the range, but range shift will ultimately result in the recruitment of juveniles at sites previously unoccupied or where recruitment had been previously unsuccessful, and so recruitment patterns must be examined. Juvenile stages tend to be more sensitive to climate stress than adult stages (Kueppers et al., 2017; Munier, Hermanutz, Jacobs, & Lewis, 2010), therefore the spatial dynamics of juveniles (i.e. saplings) both within the range and at range edges can provide an early signal to evaluate the potential for range expansion (Boisvert-Marsh et al., 2014). Recruitment patterns offer evidence of successful reproduction and establishment or, conversely, can suggest constraints, including climatic ones, on life cycle events (Smithers, North, Millar, & Latimer, 2018).

In most parts of the world, land-use and natural disturbances have modified species composition, creating opportunities for juvenile recruitment and community redistribution (Leithead, Anand, & Silva, 2010; Lembrechts et al., 2016). If natural or anthropogenic disturbances are spatially structured, their broad scale effect can be confounded with that of climate change (Lenoir, Hattab, & Pierre, 2017; Liang, Duveneck, Gustafson, Serra-Diaz, & Thompson, 2018). For instance, forestry activities in our study area (Quebec, Canada) tend to show a progression from south to north over time. Disturbance resulting from logging could match (or not) northward range shifts expected from climate change, highlighting the importance of taking disturbance into account in range shift studies. Also, climate change can modify the frequency and/or amplitude of natural (Dale et al., 2001) or human-induced disturbance (Perring et al., 2018), suggesting an important role for the interaction between climate change and disturbances in explaining range shifts (Dainese et al., 2017; Lembrechts et al., 2016). Focusing on elevational gradients,

Guo, Lenoir, and Bonebrake (2018) recently reported an interaction effect between average temperature conditions and forest loss at the site level in relation to rates of elevational shift, although the change in climate over time itself was not a major predictor in their models. The influence on range dynamics of climate change, disturbances and their interaction has not been well investigated over a broad latitudinal gradient, especially in relation to the recruitment processes that ultimately drive range shifts.

In a previous study, we addressed a gap in the climate change/biodiversity literature by providing some of the first evidence of recent latitudinal range shifts of 11 tree species at northern latitudes across a broad geographical extent (Boisvert-Marsh et al., 2014). We interpreted our results as an early signal of the potential for some species to track (or not) changes in climate, but the direct effect of climate on range shifts was not evaluated at the time. Here, we investigate patterns and processes of latitudinal range shifts of four tree species in Québec, Canada across an area ~761 100 km². We use a repeat survey of 6,309 forest plots between 1970 and 2014 and focus on common tree species that reach their northern range limit in the study area. The 15 warmest years on record have occurred consecutively since 2001 (National Center for Environmental Information, 2016) and so species could have benefited from extended growing seasons, especially at the northern edge of their distribution. We first define changes in latitudinal limits in relation to changes in tree saplings' occurrence patterns within forest plots resampled over two time periods (1970–1977 and 2003–2014). By examining changes in the frequency of occurrences in different portions of the study area along a latitudinal gradient, we can identify spatially explicit patterns of loss or gain (sapling recruitment) resulting in the shifts observed. As disturbances can provide opportunities for recruitment separately or in interaction with climate change (Leithead et al., 2010; Zhang, Bergeron, Zhao, & Drobyshev, 2015), we then estimate the probability of observing a recruitment event in response to changes in climate, disturbance and their interaction, using a multimodel selection approach. We generally expected the probability of sapling recruitment to increase with warming, especially at these high latitudes, but that relationship can change in space (Cavanaugh et al., 2014; Fisichelli, Frelich, & Reich, 2014; Miller, Wilson, Sherriff, & Walton, 2017). We also expected the disturbance signal to be more important than the climate signal given the prevalence of disturbances in the study area.

2 | MATERIALS AND METHODS

2.1 | Study area

The province of Québec (Canada) encompasses about 2% of all forested areas in the world and a significant proportion of all boreal forests (Saucier, Grondin, Robitaille, & Bergeron, 2003); *Ministry of Forests, Wildlife and Parks* (MFFP, <<https://mffp.gouv.qc.ca/the-forests/international/?lang=en>>). Our study area (Figure 1) covers roughly 761,100 km² in Quebec, from 45°N to 53°N (the northern limit of current forestry activities), and from 80°W to 61°W (Saucier

et al., 2003). Two major vegetation zones, each with two forest types, are part of the study area: the northern temperate zone and the boreal forest zone (Saucier et al., 2003). The former includes *Acer saccharum*-dominated forests to the south and mixed *Abies balsamea* (L.) Mill.-*Betula alleghaniensis* Britt. forests to the north. The latter includes *A. balsamea*-*Betula papyrifera* Marsh. forests to the south and vast *P. mariana*-moss forests to the north (Saucier et al., 2003).

2.2 | Data collection

We used extensive forest inventories of permanent plots to monitor long-term forest dynamics throughout the study area, ensuring consistency in sampling (MFFP, 2015; Dataset available from <https://www.donneesquebec.ca/recherche/fr/dataset/placettes-echantillons-permanentes-1970-a-aujourd-hui>). Data collection started in 1970 and is ongoing. Four inventory campaigns have been completed across the permanent plot network. Previous work (Boisvert-Marsh et al., 2014) considered range shifts in tree species between 1970 and 1977 (hereafter called P₁) and 1992–2002 (P₂). This study uses P₁ and the most recent complete inventory available from the MFFP, 2003–2014 (hereafter called P₃ for continuity). Plots of 400 m² are usually paired for sampling at an average distance of 425 m. Plot density decreases northward, along with tree species diversity (one pair of plots per 26 km² in the *A. saccharum* forest, one pair per 104 km² in the *Abies balsamea* forest and one pair per 259 km² in the *Picea mariana* forest).

We focused on four deciduous species typical of the temperate and mixed forest domains: *Fagus grandifolia* Ehrh., *Acer saccharum* Marsh., *Betula alleghaniensis* Britt. and *Acer rubrum* L. All species reach their northern limit in the study area, showed significant northward expansion in an earlier time period (Boisvert-Marsh et al., 2014), and had sufficient occurrences for statistical analysis. We used species data from the earliest (1970–1977; P₁) and most recent inventory (2003–2014; P₃); each inventory took between 7 and 11 years to complete. Within these inventories, the precise survey date at each plot was known and used to access climate information. Plots were sampled on average 33.8 ± 3.79 years apart. Of those plots where at least one of the study species was likely to be observed, 72% were inventoried within 3 years of the mean (minimum: 26 years, maximum: 44 years). So, we expect the temporal discrepancy across the study area to be minimal on overall forest dynamics. For inventory purposes, the MFFP defines a sapling as having a diameter at breast height (DBH) of 1.1–9.0 cm DBH, grouped by 2 cm DBH classes, and ≥1.3 m tall. Saplings of the smallest size class (1.1–3.0 cm DBH; hereafter called 'saplings') are considered to be indicative of successful establishment and were retained for this study. Sapling sampling methods are consistent between inventories, but seedling data collection only begun in P₂ and therefore seedlings were not considered. Species occurrence data, as indicated by presence/absence at a plot, were used to assess changes in spatial patterns over time. Saplings of a given species were either present or absent in P₁ and P₃. Climate has been shown to predict frequency of occurrence at

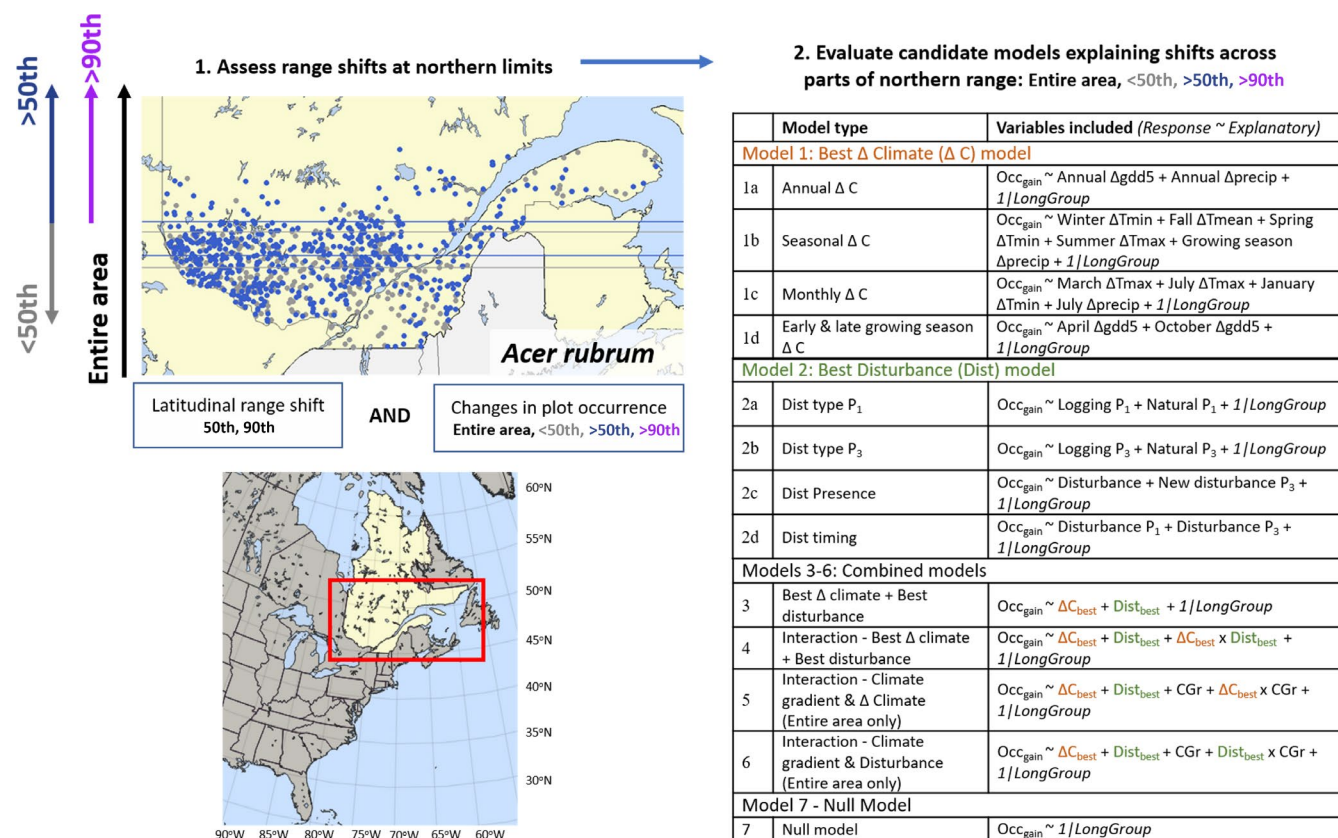


FIGURE 1 Workflow for assessing how latitudinal range shifts are related to climate change and/or disturbance in Quebec, Canada.

1) *Assess range shifts*, that is, magnitude and direction of latitudinal shifts as the change in limit location over time (*Acer rubrum* shown), lower limits = 50th; upper limits = 90th; Grey: 1970–1977 (P₁). Blue: 2003–2014 (P₃), and change in plot occurrence (occurrence in P₁; occurrence in P₃). Points show plots occupied in their respective inventories (P₁ or P₃). Range shifts are quantified as the difference between the pairs of lines. Occurrence changes were assessed for four different areas of the distribution, based on the calculated latitudinal limit in P₁ (example shown on left): Entire suitable area, south of the 50th percentile of distribution (<50th), north of the 50th percentile (>50th) and north of the 90th percentile (>90th). 2) *Evaluate candidate models explaining shifts*: We evaluated the probability of observing an occurrence gain (Occ_{gain}) in relation to climate change ($\Delta climate$), disturbance, and their interaction. Model 1 (Best ΔC) and Model 2 (Best Disturbance) were selected (by lowest AIC) each from four non-nested submodels including either climate change variables or disturbance variables. Models 3 combined the climate change variable with the highest model Wald z value from the $\Delta Climate$ model and the disturbance variable with the highest model Wald z value from the Disturbance model. Model 4 includes an interaction term between climate and disturbance and Models 5 and 6 include a climate gradient variable (CGr: growing degree days, base 5°C in P₃). For more details, see Appendix S1

different latitudes better than species importance value in a plot (Canham & Thomas, 2010; Chambers, Périé, Casajus, & Blois, 2013), especially approaching species range limits (Pironon et al., 2017).

2.3 | Disturbance data

In each plot, the presence/absence of each type of disturbance was noted during each inventory (for a list of variables, see Figure 1 and Appendix S1). When the inventories began (P₁), 29.0% of plots had been affected by natural (wind-throw, fire, insect outbreaks) and/or anthropogenic disturbances (i.e. logging: selective harvest, partially harvested or clearcut). The proportion of disturbed plots increased to 30.6% in P₃, mainly driven by anthropogenic disturbance (15.8% of plots in P₁, 19.0% in P₃), whereas natural disturbances were somewhat less prevalent in P₃ over P₁ (13.2% of plots in P₁, 11.6% in P₃). In this study, we differentiated only between 'natural disturbances'

(all types) and 'logging' (all types), and their timing as observed in each inventory.

2.4 | Climate data

Climate data used in this study cover the period from 1965 to 2014. This includes a 5-year period prior to the first forest inventory in 1970 and ends with the most recent inventory. These climate data were modelled from Environment Canada weather station data using the BioSIM program (Regnière & Saint-Amant, 2014), which interpolates temperature and precipitation around a sampling plot at a 1-km resolution using a weighted average of the four closest weather stations, correcting for differences in elevation. Climate observations attributed to the forest plots reveal a study area characterized by strong latitudinal climatic gradients. When all plots are considered over the whole time period, mean

annual temperature ranged from 6.9°C in the south to −4.4°C in the north. During that time, mean annual temperature increases ranged from +0.08°C to +1.91°C, with no plots showing a decrease. There were, however, monthly or seasonal variations, including cooling for some plots. The strongest warming occurred in the southwest and centre–north of the study area and in the recent time period (P_3). Overall annual precipitation ranged from 743 mm to 1543 mm. Precipitation gradients exist, decreasing from south to north and from east to west. There was no overall significant change in annual precipitation between 1965 and 2014, but trends varied greatly among plots. Change in total annual precipitation ranged from −268 mm to +269 mm, averaging −0.302 mm overall. For a list of all climate variables used in this study, see Appendix S1 in Supporting Information.

2.5 | Assessing latitudinal shifts

We used two methods to assess latitudinal shifts (Figure 1). First, using species occurrence data, we quantified the magnitude and direction of latitudinal shifts as the change in limit location over time. We used the percentile approach to define limits because it reduces the potential bias of extreme latitudinal data points (Lenoir, Gégout, Pierrat, Bontemps, & Dhôte, 2009). Latitudinal shifts were assessed at both the median (50th) percentile of distribution and at the 90th percentile of distribution (northernmost) using relevant methods for each (described in Appendix S1 and Boisvert-Marsh et al., 2014). Note that, because of the northern location of our study area, the 50th percentile of latitude does not reflect the median latitude of the entire north–south range of a species but the median within our study area. However, within our study area, the 90th percentile does encompass the northernmost range limit for our four study species and captures sufficient presence points for statistical analysis in the next step. Latitudinal differences were calculated between inventories ($P_3 - P_1$), with positive values indicating northward latitudinal shift and negative values southward. The significance level was set to 0.05 using nonparametric methods appropriate for non-independent samples (bootstrap resampling at the 50th percentile, pooled confidence intervals at the 90th percentile).

Because latitudinal limits are determined based on occurrence patterns at different latitudes, northward shifts over time can occur when (a) the number of plots occupied south of the limit of interest (in this case, 50th or 90th percentile of latitude) decreases more than north of that limit; (b) when the number of plots occupied north of a limit increases more than south of that limit or (c) a combination of both 1 and 2. To understand changes in latitudinal limits across the northern range, we examined changes in plot occurrence between P_1 and P_3 in different portions of the study area as determined by latitudinal limits in P_1 . Three portions of the study area were evaluated: (a) south of the species' P_1 distribution median (i.e. <50th), (b) north of the species' P_1 median (i.e. >50th) and (c) north of the species' P_1 northernmost limit (i.e. >90th). McNemar's test for paired samples was used to see if the frequency of the presence/

absence in an area was statistically different ($p \leq 0.05$) between the two inventories.

2.6 | The relationship between climate change, disturbances and sapling recruitment

For this analysis, we focused on plots where a given species (saplings) was absent in P_1 , but present in P_3 . This was considered a recruitment event or occurrence gain (hereafter called Occ_{gain}). Sampled plots with Occ_{gain} were compared with plots where there was no recruitment (species absent in both time periods). Plots where there was no observed recruitment needed nevertheless to be suitable for the species. To determine suitable 'absences', we used a consensus species distribution modelling approach based on overall climate gradients related to temperature and precipitation, edaphic variables and topographic factors (summarized in Appendix S1, following methods detailed in Beauregard & de Blois, 2014, Périé & de Blois, 2016).

Climate versus disturbance—For each species, we evaluated the probability of observing an occurrence gain in relation to climate change, disturbance and their interaction. Climate change was defined as the climate conditions observed in P_3 minus the climate conditions in P_1 (hereafter called $\Delta\text{climate}$). Thirteen Δ climate variables were selected in preliminary analyses (Figure 1) and grouped based on temporal variation (annual, seasonal, early or late season, monthly). Climate variables with a finer temporal resolution may not only have as much or more weight in models as annual ones but could also point to specific life events (i.e. coincide with timing of flowering or establishment). Because we were measuring trends in climate associated with a given plot, each climate variable was averaged over a 5-year period preceding the year of a recorded observation (e.g. 2000–2005 for a 2005 observation). Disturbance was characterized by whether a plot was disturbed or not, its type (logging vs. natural), and the timing of the disturbance event ($\Delta\text{climate}$ and disturbance variables listed in Figure 1).

We used a multimodel selection approach (Alofs & Jackson, 2015) to evaluate the probability of observing an Occ_{gain} in relation to our explanatory variables. Seven candidate models, relating different combinations of climate, climate change, and disturbance to observed Occ_{gain} were tested in four different areas of the suitable distribution, based on the calculated latitudinal limit in P_1 : (a) entire suitable area, (b) south of the 50th percentile (<50th), (c) north of the 50th percentile (>50th) and (d) north of the 90th percentile (>90th). Models 1 and 2 were the best $\Delta\text{climate}$ model and the best disturbance model, respectively, selected from pre-analyses on four non-nested submodels ($\Delta\text{climate}$ 1a–1d; Disturbance – 2a–2d). Best models were selected using the lowest AIC in each subgroup. The best $\Delta\text{climate}$ variable from Model 1 and the best disturbance variable from Model 2 were identified based on the highest absolute Wald z value, as obtained from the model outputs (Appendix S2) and were then used to construct Models 3–6. By using the model Wald z value, binary and continuous variables can be compared for their relative explanatory power in the various models (Agresti, 2013).

TABLE 1 Best variable retained from the Δ climate model (Model 1) and the disturbance model (Model 2) between P_1 (1970–1977) and P_3 (2003–2014). The best variable was obtained from pre-analyses which tested models in four different areas of the distribution, based on the calculated latitudinal limit in P_1 : entire suitable area, south of the 50th percentile of distribution, north of the 50th percentile and north of the 90th percentile. The sign of the coefficient of the explanatory variable in each model indicates the relationship between the explanatory variable and occurrence gains (plots not occupied by saplings in P_1 but occupied in P_3 ; +: positive, -: negative). Variables in bold indicate the Δ climate or disturbance variable with the highest Wald z value. For complete model details (coefficient values, standard errors and p values), see Appendix S2. Species are ordered from left to right from the most meridional to most boreal northernmost limit (90th percentile) in P_1

	<i>Fagus grandifolia</i>		<i>Acer saccharum</i>		<i>Betula alleghaniensis</i>		<i>Acer rubrum</i>	
	Δ Climate	Disturbance	Δ Climate	Disturbance	Δ Climate	Disturbance	Δ Climate	Disturbance
Entire suitable	October Δ gdd5 (+) ^{**}	Natural P_1 (-) ^{**}	Annual Δ gdd5 (-)	Natural P_3 (-)	Summer Δ TMax (+) ^{**}	Logging P_1 (+) ^{**}	October Δ gdd5 (-) ^{**}	Logging P_1 (-) ^{****}
<50th	April Δ gdd5 (-) ^{***}	Logging P_1 (-)	Annual Δ gdd5 (-) ^{***}	New Disturbance P_3 (+)	April Δ gdd5 (-) ^{**}	Logging P_1 (+) [*]	October Δ gdd5 (-) [*]	Natural P_1 (+) ^{**}
>50th	October Δ gdd5 (+) [*]	Natural P_1 (-) ^{**}	April Δ gdd5 (+) ^{**}	Logging P_3 (+)	April Δ gdd5 (+) [*]	Natural P_1 (-) [*]	October Δ gdd5 (-) [*]	Logging P_1 (-) ^{***}
>90th	July Δ precip (+) ^{****}	Disturbance (-)	April Δ gdd5 (+)	Natural P_3 (-)	Annual Δ gdd5 (-)	Logging P_1 (-)	October Δ gdd5 (-) [*]	Logging P_1 (-) [*]

Note. Significance levels: ^{*} $p < 0.05$; ^{**} $p < 0.01$; ^{***} $p < 0.001$; ^{****} $p < 0.0001$.

Models 3–6 evaluated combinations of best Δ climate and best disturbance variables, as main effects and in interaction, including their interactions with the climate gradient (Figure 1, best variables listed in Table 1). Logistic regression with a logit link using a mixed-model approach was used to compare plots with Occ_{gain} to plots without gains. Longitudinal band (2° wide) was used as a random factor in the generalized linear mixed models to account for potential variation in the response variable induced by the broad east–west climatic gradient. All data analyses were done in R, version 3.3.2 (R Core Team, 2016).

3 | RESULTS

3.1 | Latitudinal shifts

The latitudinal limits of saplings of all four species shifted northward between P_1 and P_3 for both the 50th and 90th percentiles of distribution (Figure 2a). These shifts, however, were driven by different patterns of occurrence changes depending on the species and the portion of the study area examined (Figure 2b). The greatest latitudinal shifts across the study area were seen for *A. rubrum* followed by *A. saccharum*. Their limits moved significantly northward for both the 50th and 90th percentiles (*A. rubrum*: +27.1 km and +23.6 km; *A. saccharum*: +18.2 km and +17.4 km; Tables S1.1 and S1.2). *Acer rubrum* underwent significant occurrence increases relative to the P_1 in all parts of the study area, but more so north of the 50th, including north of the 90th (Figure 2b). *Acer saccharum*, on the other hand, underwent significant occurrence decreases south of the 50th, but saw significant occurrence increases north of the 50th and 90th percentiles. Smaller occurrence increases were observed compared to those of other species, but the combined effect of increase in the north and decrease in the south contributed to amplify latitudinal shifts for this species (Figure 2b).

The latitudinal range of *B. alleghaniensis* shifted the least of all species, and its shifts were non-significant (+1.5 km and +4.2 km at 50th and 90th percentiles respectively; Figure 2a, Tables S1.1 and S1.2). Nevertheless, *B. alleghaniensis* showed significant occurrence increases across the study area. The fact that these occurrence increases were relatively similar in all portions of the study area explains the limited overall shifts observed. *Fagus grandifolia* showed a non-significant northward shift at the 50th (+2.3 km, Table S1.1), but a significant northward shift at the 90th (+13.2 km, Table S1.2). Significant, relatively high occurrence increases were observed across the study area, but more so north of the 90th, explaining the significant shift observed at this latitude.

3.2 | The relationship between climate change, disturbances and sapling recruitment

Observed occurrence gains (Figure 3) and their association with climate change and disturbance (Figure 4) were spatially structured across the study area. When considering Model 1 alone (Δ climate), predictors measuring changes in early and late growing season

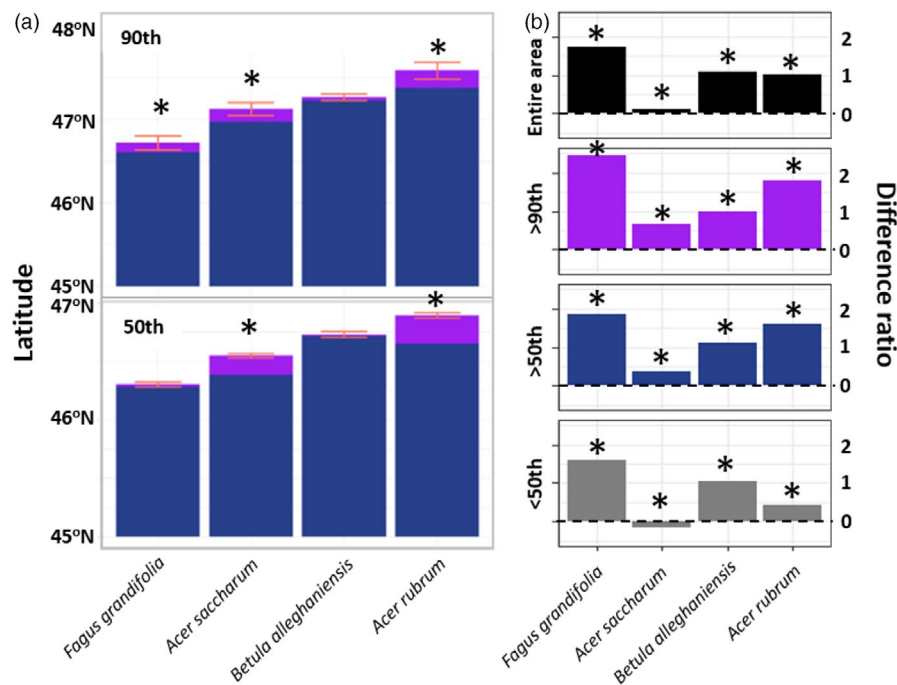


FIGURE 2 Summary of evidence for latitudinal shifts between P_1 (1970–1977) and P_3 (2003–2014) for saplings of four tree species in Quebec, Canada using latitudinal limit shifts (2a) and occurrence changes (2b). For latitudinal limit shifts, the latitudinal limit in P_1 is indicated by the height of the blue bar while the P_3 limit and its difference relative to the P_1 limit is indicated by the height of the purple bar. This was assessed at both the median (50th, bottom) and the northernmost (90th, top) limit. For changes in plots occupied, the difference in number of plots occupied between the two inventories relative to the number occupied in P_1 is represented for four areas (Entire suitable area, south of the 50th percentile of distribution (<50th), north of the 50th percentile (>50th), and north of the 90th percentile (>90th)). *denotes significant latitudinal shifts or change in number of plots occupied ($p < 0.05$)

variables were most often retained in pre-analyses: October $\Delta gdd5$ (six times), April $\Delta gdd5$ (five), Annual $\Delta gdd5$ (three) and twice for monthly variables (Summer $\Delta Tmax$: one and July $\Delta precipitation$: one) (Table 1). When considering Model 2 only, which compared disturbance variables, those most often retained in pre-analyses were the models distinguishing between human or natural disturbance in the earliest time period (11 times—logging P_1 : 7, natural P_1 : 4) or the latest (three times—natural P_3 : 2, logging P_3 : 1), followed by presence of disturbance (1) and newly disturbed in P_3 (1) (Table 1).

For forest plots, although annual $\Delta gdd5$ has increased almost everywhere, this increase has been more pronounced in the north-western portion of the study area (Figure 4d). Maximum summer temperatures have increased (Figure 4c) and precipitation (Figure 4e) has decreased to greater extents in the west. October $\Delta gdd5$ shows positive (i.e. longer growing season in the fall) values mostly in the centre of the study area and around the Gaspé peninsula in the east, but negative ones elsewhere (Figure 4b). April $\Delta gdd5$ shows the strongest increases in the southeastern corner (longer growing season in the spring) along the American border (Figure 4a).

Overall, among all species, the best models (Table 2) were (in decreasing order): Model 3 ($\Delta C_{best} + Dist_{best}$ —seven times), Model 1 ($\Delta climate$ only—two), Model 4 ($\Delta C_{best} \times Dist_{best}$ interaction—two), Model 5 ($Cgr \times Dist_{best}$ interaction—two), Model 2 (Disturbance only—one), Model 6 ($Cgr \times \Delta C_{best}$ interaction—one) and Model 7 (null model—one). For individual species, however, relationships

between the predictors and response varied depending on the portion of the study area. For *F. grandifolia*, larger increases in October $\Delta gdd5$ had a positive impact on recruitment when the entire suitable area was considered. This was also true north of the 50th percentile, even though half of the suitable plots saw decreases in October $\Delta gdd5$ (Figure 4b). At the northern edge (>90th percentile) where this species had its strongest occurrence increases (Figure 2b), the best model included $\Delta climate$ only, and larger increases in July $\Delta precipitation$ were associated with higher probability of gains in sapling occurrence (Occ_{gain}). Other seasonal climate variables at this latitude tended to have positive effects, except for an increase in March $\Delta minimum$ temperature. South of the 90th percentile, larger increases in April $\Delta gdd5$ negatively affected Occ_{gain} . When retained in the model, logging in P_1 had a negative impact on Occ_{gain} in all portions of the study area, but was consistently outscored by $\Delta Climate$ variables in the best models. Logging in P_1 tended to interact positively with an extended growing season in the spring south of the 50th percentile, but the relationship was only marginally significant.

Acer saccharum's response was spatially structured across the entire study area, as evidenced by the inclusion of the latitudinal climate gradient term ($gdd5 P_3$ (CGr): Table 2). South of the 50th percentile, where occurrence decreases were observed (Figure 2b), the best model was $\Delta climate$ only (Model 1) and larger increases in annual $\Delta gdd5$ reduced the probability of observing an occurrence gain. Approaching the northern range limit, models including ΔC_{best} and

$\text{Dist}_{\text{best}}$ had the most support, with larger increases in April Δgdd5 increasing the probability of Occ_{gain} . In these models, disturbance variables for the most recent period (P_3) were retained.

Betula alleghaniensis had similar occurrence increases across the study area (Figure 2b). Larger increases in maximum summer temperatures increased the probability of Occ_{gain} overall, while in the south larger increases in April Δgdd5 had a negative impact. North of the 50th percentile, a disturbance only model was retained, with natural disturbances in P_1 having a marginally negative impact. The null model was selected north of the 90th.

Finally, *A. rubrum* had significant gains everywhere, but not as great in the southern portion of the study area (Figure 2b). This species seemed most impacted by disturbances relative to climate (Table 2), with past natural disturbances increasing the probability of Occ_{gain} south of the 50th percentile, while logging reduced gain probability north of the 50th. Extending the growing season in the fall reduced Occ_{gain} probability, most significantly at the northernmost edge.

4 | DISCUSSION

The latitudinal shift patterns we observed for four northern temperate hardwood tree species were generally consistent with climate change predictions of northward shifts at the northern range limit (Périé & de Blois, 2016) and with those of previous studies (Boisvert-Marsh et al., 2014; Sittaro et al., 2017). They also reveal shifts that result from spatially explicit responses, with different combinations of climate change and disturbance variables influencing the probability of sapling occurrence gains depending on location within the study area. Moreover, the best climate models retained mostly temperature rather than precipitation variables, and monthly or seasonal ones over annual ones. The former was expected as we focused on latitudinal gradients where strong temperature gradients prevail. The latter suggests the influence of local climate change at times critical for life events (reproduction, dispersal, establishment) leading to successful recruitment. Overall, our findings show responses to warming that varies depending on the geographical position and warming intensity.

We generally expected the disturbance signal to be greater than the climate signal given the prevalence of disturbances in the study area. On the contrary, support for climate change variables mostly outweighed (based on absolute Wald z value) disturbance ones. South of our study area into the United States, climate gradients and their changes were also shown to have impacts on tree range shifts (Bose, Weiskittel, Wagner, & Pauchard, 2017a; Fei et al., 2017). Interestingly, despite our studying tree species at their northern range limit in eastern North America, greater extension of the growing season did not always lead to a higher probability of sapling recruitment at a site. In fact, warming may be becoming stressful at southern locations. This was particularly evident for *F. grandifolia* and *A. saccharum*, the species with the most meridional northern limits in our dataset, and also to a lesser degree, for

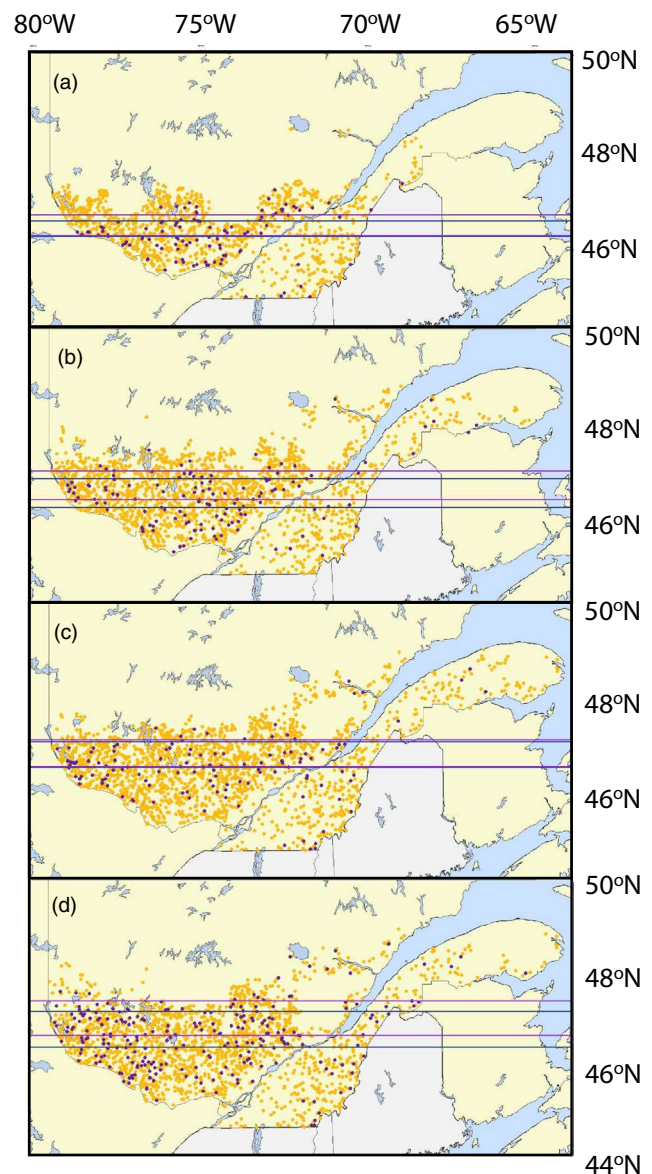


FIGURE 3 Maps of sapling occurrence gains for four tree species between P_1 (1970–1977) and P_3 (2003–2014) in Quebec, Canada. Species, ordered from most meridional to most boreal northernmost limit (90th percentile) are (a) *Fagus grandifolia*, (b) *Acer saccharum*, (c) *Betula alleghaniensis* and (d) *Acer rubrum*. The blue lines represent the 50th and 90th percentiles of latitude in P_1 . The purple lines represent the 50th and 90th percentiles of latitude in P_3 . Purple dots represent a sapling gain observed in P_3 while yellow dots represent plots suitable for a given species, based on species distribution models

B. alleghaniensis. Conversely, the relationship between warming or extension of the growing season and sapling occurrence gains tended to be positive at northern locations for *A. saccharum* and *F. grandifolia*, coinciding with northward expansion. South of our study area, in the United States, *A. rubrum* and *B. alleghaniensis* showed northward shifts in abundance over a similar time period but, in contrast to our study, *A. saccharum* and *F. grandifolia* showed southward shifts (Fei et al., 2017). If the southward shift

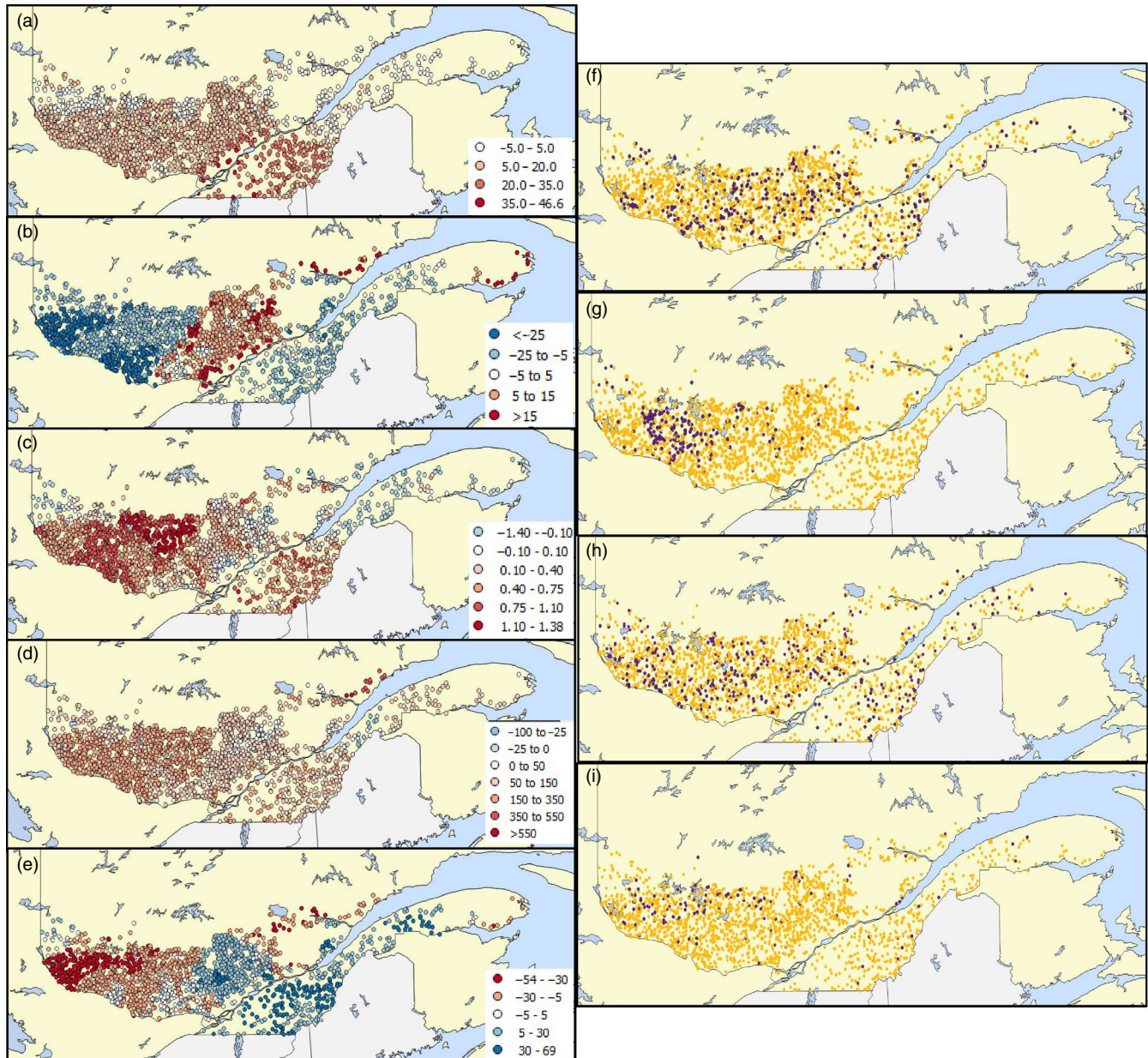


FIGURE 4 Maps of changes in selected climate change variables (Δ climate: (a–e) and disturbance (f–i) between P_1 (1970–1977) and P_3 (2003–2014) in Quebec, Canada. The Δ climate variables depicted are the ones selected in Δ climate models (Model 1): from top to bottom, (a) April Δ gdd5 (change in growing degree days), (b) October Δ gdd5, (c) Summer maximum temperature (Summer Δ TMax), (d) Annual Δ gdd5, (e) Δ July precipitation). Legends indicate the magnitude of change; blue denotes plots that experienced cooler temperature ($^{\circ}$ C), fewer growing degree days or wetter conditions (mm) between the two periods and red denotes plots that experienced warmer temperature, more growing degree days or drier conditions. The disturbance variables depicted are those selected in disturbance models (Model 2): from top to bottom, (f) Logging P_1 , (g) Natural P_1 , (h) Logging P_3 , (i) Natural P_3 . Purple dots represent plots where disturbance and its type were recorded in a given period while yellow dots represent plots where no disturbance was recorded

they observed for *A. saccharum* results from declining abundance in the northern part of their study area (Fei et al., 2017), this could be consistent with declining occurrence reported in the south of our study area and into New England (Bishop et al., 2015; Bose, Weiskittel, & Wagner, 2017b). On the other hand, reports of increasing occurrences in the south of our study area and through New England for *F. grandifolia* would contrast with southward shifts in abundance reported by Fei et al. (2017). Combining

insights across the range could be useful to contextualize results in specific areas, although the time periods under consideration may not be similar. Divergence in the responses of early-life stages across large latitudinal gradients has also been shown for two European congeneric species of *Acer* (*A. pseudoplatanus* L. and *A. platanoides* L.). Seedling germination and survival of these two species responded differently to experimental warming and precipitation treatments depending on their provenance along a climate

TABLE 2 Best models from multimodel selection assessing the relationship between occurrence gain compared to climate change and disturbance. For each species, candidate models were assessed for four different areas of the distribution, based on the calculated latitudinal limit in P_1 (1970–1977): entire suitable area, south of the 50th percentile of distribution (<50th), north of the 50th percentile (>50th) and north of the 90th percentile (>90th). The variable with the highest overall model Wald z value is in bold. The coefficient sign is indicated (+: positive relationship between occurrence gains and explanatory variables; -: negative relationship). The difference between the AIC_c score of the best model and null model (ΔAIC_c), Akaike weight (Wt) and R^2 of the best model are included for reference. For complete model outputs, see Appendix S2

	Best model	AIC_c	ΔAIC_c w/null (Wt)	R^2	Explanatory variables
<i>Fagus grandifolia</i>					
Entire suitable	Model 3 $\sim \Delta C_{best} + Dist_{t_{best}}$	700.16	-4.35 (0.30)	0.0732	October $\Delta gdd5$ (+) + Natural P_1 (-)
<50th	Model 4 $\sim \Delta C_{best} + Dist_{t_{best}} + \Delta C_{best} \times Dist_{t_{best}}$	287.36	-3.38 (0.39)	0.1204	April $\Delta gdd5$ (-) + Logging P_1 (-) + April $\Delta gdd5 \times$ Logging P_1 (+)
>50th	Model 3 $\sim \Delta C_{best} + Dist_{t_{best}}$	405.85	-5.9 (0.55)	0.0847	October $\Delta gdd5$ (+) + Logging P_1 (-)
>90th	Model 1 $\sim C$ (Monthly ΔC)	319.23	-4.46 (0.41)	0.1193	January $\Delta TMin$ (+) + March $\Delta TMax$ (-) + July $\Delta TMax$ (+) + July $\Delta Precip$ (+)
<i>Acer saccharum</i>					
Entire suitable	Model 5 $\sim \Delta C_{best} + Dist_{t_{best}} + CGr + \Delta C_{best} \times CGr$	935.68	-6.06 (0.56)	0.0504	Annual $\Delta gdd5$ (+) + Natural P_3 (-) + $gdd5 P_3$ (+) + Annual $\Delta gdd5 \times gdd5 P_3$ (-)
<50th	Model 1 $\sim \Delta C$ (Annual ΔC)	304.61	-7.69 (0.71)	0.1612	Annual $\Delta gdd5$ (-) + Δ Total Annual Precipitation (-)
>50th	Model 3 $\sim \Delta C_{best} + Dist_{t_{best}}$	625.37	-4.48 (0.49)	0.0403	April $\Delta gdd5$ (+) + Logging P_3 (+)
>90th	Model 4 $\sim \Delta C_{best} + Dist_{t_{best}} + \Delta C_{best} \times Dist_{t_{best}}$	247.19	-1.1 (0.36)	0.035	April $\Delta gdd5$ (+) + Natural P_3 (+) + April $\Delta gdd5 \times$ Natural P_3 (-)
<i>Betula alleghaniensis</i>					
Entire suitable	Model 6 $\sim \Delta C_{best} + Dist_{t_{best}} + CGr + Dist_{t_{best}} \times CGr$	890.88	-7.56 (0.24)	0.0646	Summer $\Delta TMax$ (+) + Logging P_1 (+) + $gdd5 P_3$ (-) + Logging $P_1 \times gdd5 P_3$ (+)
<50th	Model 3 $\sim \Delta C_{best} + Dist_{t_{best}}$	410.03	-4.32 (0.55)	0.0859	April $\Delta gdd5$ (-) + Logging P_1 (+)
>50th	Model 2 $\sim Dist$ (Disturbance type P_1)	485.98	-1.4 (0.37)	0.0415	Logging P_1 (+) + Natural P_1 (-)
>90th	Model 7 Null	151.99	0 (0.48)	—	—
<i>Acer rubrum</i>					
Entire suitable	Model 6 $\sim \Delta C_{best} + Dist_{t_{best}} + CGr + Dist_{t_{best}} \times CGr$	1,410.24	-28.89 (0.93)	0.0738	October $\Delta gdd5$ (-) + Logging P_1 (-) + $gdd5 P_3$ (-) + Logging $P_1 \times gdd5 P_3$ (+)
<50th	Model 3 $\sim \Delta C_{best} + Dist_{t_{best}}$	544.62	-3.76 (0.41)	0.0276	October $\Delta gdd5$ (-) + Natural P_1 (+)
>50th	Model 3 $\sim \Delta C_{best} + Dist_{t_{best}}$	873.61	-18.72 (0.54)	0.0751	October $\Delta gdd5$ (-) + Logging P_1 (-)
>90th	Model 3 $\sim \Delta C_{best} + Dist_{t_{best}}$	281.05	-4.7 (0.53)	0.0729	October $\Delta gdd5$ (-) + Logging P_1 (-)

Note. P_1 —1970–1977; P_3 —2003–2014. ΔC_{best} — Δ climate variable retained for a given species $\times$ area of range considered. $Dist_{t_{best}}$ —Disturbance variable retained. CGr—Climate gradient across study area, as denoted by $gdd5 P_3$. Significance levels: $p > 0.1$; $p > 0.05$; $p > 0.01$; $p > 0.001$; $p > 0.0001$; $p > 0.00001$.

gradient (Carón et al., 2015). Divergent latitudinal responses with climate warming have also been reported for growth of *Picea mariana*, suggesting that it could better withstand warming at northernmost locations than at southern locations (D'Orangeville et al., 2016). These enhanced growth effects at the northern range limit, however, could be transitory for this species as warming continues (D'Orangeville et al., 2018), especially for juvenile stages which are more sensitive to climate stress than adult stages (Kueppers et al., 2017; Munier et al., 2010).

Identifying responses to specific climate variables across broad latitudinal gradients can point to biological mechanisms and their role in facilitating or hampering species migration or adaptation (De Frenne et al., 2013). For instance, changing spring conditions put species at risk when late season frosts interrupt phenological processes (Körner & Basler, 2010). Frost damage is an increasingly common occurrence under climate change (Augsburger, 2013; Liu et al., 2018), and could explain in part why increasingly earlier springs can negatively impact recruitment at southern locations where earlier budburst and/or seedling germination would be vulnerable to spring frosts (Burns & Honkala, 1990; Hufkens et al., 2012). At northern locations, fruit maturation and germination could benefit from a fall extension of the number of growing degree days (Morin, Augspurger, & Chuine, 2007). Over broad ranges, divergent responses for a given species may indicate intraspecific adaptation to local climatic conditions, resulting in differentiation of populations within the broad climatic tolerance of a species (Davis & Shaw, 2001). These adaptations likely evolved in fairly stable climate conditions over the last few millennia, but the unprecedented, current rate of climate change could favour rapid migration over adaptation for long-lived species (Ordonez & Williams, 2013). A close link between climate and phenology has implications for management, such as assisted migration efforts and seed transfers between climate zones (Pedlar et al., 2012).

When disturbance was retained in selected models, past logging had a negative effect on *F. grandifolia* and *A. rubrum*, the latter being the species most significantly affected by disturbance compared to climate. However, logging had a positive impact on *B. alleghaniensis*, possibly because of its commercial importance, resulting in it being selected during forest management activities. Management legacies have shown to leave lasting imprints in the European temperate forest (Perring et al., 2018), in the boreal forest of eastern Canada (Cyr, Gauthier, Bergeron, & Carcaillet, 2009) and in the eastern United States (Nowacki & Abrams, 2015), but the trajectories are not always well understood. For instance, fire suppression has been proposed as a reason for *A. rubrum*'s expansion throughout its range (Abrams, 1998), but this species has also increased at other locations, favoured mainly by smaller scale disturbances over larger ones (Fisichelli et al., 2014; Leithead et al., 2010).

Disturbances could act synergistically with climate warming, opening the canopy, reducing competition locally and facilitating species turnover at suitable sites (Thom, Rammer, & Seidl, 2017). Disturbances could also have the opposite effect in areas where warming thresholds for recruitment are reached, exacerbating the

negative impact of climate change. In our study area, although the model comprised of an interaction term between climate change and disturbance (Model 4) competed at times with the best selected model, this model was only selected twice with the interaction term being otherwise non- or weakly significant. For *A. saccharum* north of the 90th percentile, spring warming and recent natural disturbances interacted negatively, possibly limiting range expansion, whereas for *F. grandifolia* in the south, spring warming interacted positively with logging. It is possible that for some species, changes in the disturbance regime, including fire, may limit their ability to migrate and persist in the boreal forest (Boulanger, Taylor, Price, Cyr, & Sainte-Marie, 2018). Using physiologically based simulations in the northeastern United States, Liang et al. (2018) reported that disturbances may expedite species' recruitment into new sites, but they had little effect on the velocity of simulated range boundary shifts. Following fires in California that occurred since 2004, Young et al. (2019) found little evidence of disturbance-mediated vegetation community reorganization on range shifts, from either low sensitivity of species to post-disturbance weather or widespread recruitment declines precipitated by unusually dry conditions. Interactions between climate change and disturbances warrant further investigation as management practices will need to be adapted to a changing climate, particularly to facilitate migration.

The influence of climate change, disturbances and their interactions can be much more complex than what we have captured in our models. Climate extremes are likely to be much more consequential for plants than changes in mean climate variables (Zimmermann et al., 2009). Vegetation responses could lag in time and disturbances can have long legacies, making it hard to assess the appropriate temporal scale for evaluating their relative impact on range dynamics (Boucher, Grondin, & Auger, 2014; Kohyama & Shigesada, 1995; Renwick & Rocca, 2015). Moreover, edaphic conditions play also an important role in limiting distribution (Beauregard & de Blois, 2014; Brown & Vellend, 2014), but we included them in selecting suitable habitats (Appendix S1). Land-use and landscape fragmentation also influence the ability of plant species to reach suitable sites (Miller & McGill, 2018) and this influence may be more important where human activities are concentrated. Altered nutrient cycling could favour certain tree species to the detriment of others. For example, even if the trend has reversed since (Houle, Marty, & Duchesne, 2015), increased nitrogen availability before the start of P₃, caused by atmospheric nitrate deposition (Fenn et al., 1998), could hasten turnover from coniferous to deciduous stands. It may have favoured the establishment of N-responsive species such as *A. saccharum* (Rothstein, Zak, & Pregitzer, 1996). Finally, the observed patterns of range shift do not imply that the species will keep up with the rapidly changing climate (Aubin et al., 2018; Boisvert-Marsh et al., 2014; Sittaro et al., 2017), although trees' recent latitudinal shifts match the direction predicted by species distribution models (Périé & de Blois, 2016). In spite of these limitations and given the paucity of data on the relationship between ongoing climate change, disturbances and range dynamics, our models offer significant insights on forests' response.

In summary, tree recruitment response to climate change is spatially explicit, depending on the portion of the study area examined. Even at these northern latitudes, there is evidence of population differentiation and varying spatial patterns of climate change, with positive effects of warming on recruitment for a given species towards its northern range edge and negative ones in the southern portion of the study area. The importance of early- or late-season climate variables in explaining recruitment trends suggests that species are affected at critical stages of their life cycle for reproduction and/or dispersal. Given the prevalence of anthropogenic disturbances in the study area, climate change's influence on recruitment is stronger than expected, and responses to disturbances tend to be species-specific. There is limited support for interaction effects between climate change and disturbances in this dataset. Continued monitoring of forest species is crucial to understand fully the long-term implications of these spatially explicit responses to global changes and the factors that could influence the capacity of species to migrate or adapt, with the goal of adapting management and conservation practices accordingly.

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AUTHOR CONTRIBUTIONS

L.B.-M., C.P. and S.d.B. conceived the original study idea and designed the methodology; L.B.-M. along with S.d.B. led the analysis of the data and the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

DATA ACCESSIBILITY

Forest inventory data, including species data and edaphic and disturbance variables, are available from <https://www.donneesquebec.ca/recherche/fr/dataset/placettes-echantillons-permanentes-1970-a-aujourd-hui>. The BioSIM program used to interpolate climate data is available from <https://cfs.nrcan.gc.ca/publications?id=34818>.

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